

Attentional Extremes: Examining Interference Control in ADHD Using a Picture-Word Stroop Task

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Abstract (Written by: Grace Borgerding and Soumil Kalra; edited by: Yutong Han)

The capability to stay attentive to the required goal stimulus and ignore distracting information (interference control) is vital for effective decision making. Interference control is arguably even more critical when making timely decisions under conflicting information such as on a Picture Word Stroop task. This study investigated the effects of interference control on individuals reporting self-reported symptoms of attention deficit hyperactivity disorder (ADHD) versus those reporting absence of symptoms. A web-based computerized Picture Word Stroop task was presented to both groups. Participants first were presented with a cue for either a picture or a word they should focus on. They were then shown the context stimuli, a picture with a word overlaid, then judged whether the upcoming test stimuli was related conceptually to the focused aspect of the context display, while ignoring the irrelevant information that could be tested. We predicted that participants who reported ADHD symptoms would exhibit larger interference, shown through slower reaction times and lower accuracy for interference trials than for control trials. After removing identical trials, eliminating practice trials, removing the participant who stated 'prefer not to say' for ADHD status, and implementing the 75% accuracy threshold, 11 participants remained. No significant effects of reaction time interference were found. Reaction times did not support our hypothesis of larger interference effect for individuals reporting ADHD symptoms. Although accuracy results were descriptively in the hypothesized direction, showing slightly greater accuracy loss for individuals reporting ADHD, results were not statistically significant. The present study thus provided little behavioral evidence for increased interference for those with ADHD, using our short online version of the Picture Word Stroop. Nonetheless, the current findings contribute to the literature in regard to effective task design, data quality issues, and attentional control measurement for small web-based samples. Current findings were discussed within the framework of prefrontal control, anterior cingulate involvement in conflict monitoring, default mode network suppression, and some evidence from prior psychopathy research demonstrating less interference.

Attentional Extremes: Examining Interference Control in ADHD Using a Picture-Word Stroop Task

Introduction (Written by: Grace Borgerding and Soumil Kalra; edited by: Yutong Han)

Decision making depends partly on the ability to allocate attention efficiently under time constraints. In everyday life, people rarely make decisions in perfectly quiet or distraction free environments. Instead, they usually have to focus on the information that matters while suppressing information that is present but irrelevant. This process is central to cognitive control, which refers to the ability to guide thought and behavior according to current goals rather than allowing responses to be driven automatically by the most noticeable stimulus in the environment. Neurocognitive models of control often emphasize the prefrontal cortex as a system that maintains task goals and biases processing toward goal relevant information (Miller & Cohen, 2001). When conflict occurs between a goal relevant stimulus and an irrelevant distractor, the anterior cingulate cortex is thought to help monitor that conflict and signal the need for stronger control (Botvinick et al., 2001; Bush et al., 1999).

One of the most widely used ways to study interference and cognitive control is the Stroop task. In the original Stroop task, participants were asked to respond to one dimension of a stimulus while ignoring another competing dimension. Stroop (1935) showed that people are slower when the word meaning conflicts with the required color naming response, such as when the word “red” appears in blue ink. Later reviews have shown that the Stroop effect is one of the most reliable demonstrations of interference in attention and response selection (MacLeod, 1991). The task is useful because it creates a conflict between automatic processing and goal directed responding. Reading is highly practiced and automatic, so it interferes with the less automatic task of naming ink color.

Modified Stroop tasks extend this logic beyond color naming. The Picture Word Stroop task requires participants to attend to one part of a display while ignoring another. Earlier work by Gernsbacher and Faust (1991) used picture word relationships

to study suppression during comprehension, and later studies adapted this logic to examine selective attention in psychopathy (Hiatt et al., 2004; Newman et al., 1997; Vitale et al., 2007). In this version of the task, a participant sees a context display containing a picture and a word. Depending on the cue, the participant must attend to either the picture or the word. A later test stimulus appears, and the participant decides whether that test stimulus is conceptually related to the attended component of the earlier display. The ignored stimulus can create interference when it is related to the test stimulus even though the attended stimulus is not. For example, if the participant is told to attend to the picture, they might see a picture of a baseball player with the word “rain” superimposed on it, followed by a test picture of an umbrella. The correct answer would be “unrelated” because umbrella is not related to baseball player, but the ignored word “rain” is related to umbrella. If the participant automatically processes the ignored word, the distractor may slow the response or increase the likelihood of an incorrect “related” response.

This task is particularly interesting for understanding clinical differences in attention because it dissociates goal directed attention from automatic processing of distracters. Sustained attention, inhibition, and goal maintenance may be impaired in ADHD and prevent the suppression of irrelevant information (Barkley, 1997; Klein et al., 2025), making those with ADHD particularly susceptible to interference because the bottom up distractors are in greater competition with the top down goal set. This general hypothesis is supported by research in neuroscience that has demonstrated that people with ADHD have different activation in frontostriatal and frontoparietal control networks, poorer engagement of inhibition and conflict monitoring areas, and a decreased ability to suppress the default mode network when processing external information (Castellanos et al., 2008; Fassbender et al., 2009; Rubia, 2018; Rubia et al., 1999). This may explain why ADHD is associated with increased difficulty in tasks that require an individual to continue to hold their goal in mind while they simultaneously filter distracters.

Psychopathy research offers an interesting contrast. Prior Picture Word Stroop studies have found that individuals high in psychopathy can sometimes show reduced

interference from task irrelevant information (Hiatt et al., 2004; Newman et al., 1997; Vitale et al., 2007). This pattern has been interpreted through the response modulation hypothesis, which proposes that psychopathy involves an abnormal tendency to remain locked onto dominant goal relevant information while failing to process peripheral cues that are outside the current focus of attention (Newman et al., 1997). In other words, while ADHD may involve too much distractor influence, psychopathy research suggests that some individuals may show too little processing of peripheral information. Although the present study did not directly recruit a psychopathy group, prior psychopathy findings provide a useful literature based comparison point for understanding how different attentional profiles can produce opposite patterns on Stroop based tasks.

The present study used a web based Picture Word Stroop task to compare interference effects in participants with and without self reported ADHD. Participants completed practice trials followed by main task trials in which they attended either to the picture or the word, then judged whether a test stimulus was related or unrelated to the attended component of the context display. We measured reaction time and accuracy across experimental interference trials and comparison trials. We hypothesized that participants with self reported ADHD would show a larger interference effect than non ADHD controls, reflected by slower reaction times and lower accuracy on experimental interference trials compared with comparison trials.

Methods (Written by: Soumil Kalra; edited by: Grace Borgerding and Yutong Han)

Participants

Participants completed the study online through a web-based Picture Word Stroop task. The raw dataset contained 399 rows of trial level data. Because several participant blocks were accidentally duplicated in the exported spreadsheet, exact duplicate rows were removed before analysis. After duplicate removal, the cleaned dataset contained 315 unique rows from 15 participants. Each participant completed 6 practice trials and 15 main trials, giving 21 trials per participant.

Participants self-reported ADHD status at the beginning of the task. After duplicate removal, the sample included 10 participants who reported no ADHD diagnosis, 4 participants who reported an ADHD diagnosis, and 1 participant who selected “prefer not to say.” Because the main research question compared ADHD and non ADHD participants, the participant who selected “prefer not to say” was excluded from group based analyses. This left 14 participants for initial ADHD group comparison. As an additional data quality criterion, participants with less than 75 percent accuracy on the main task were excluded from the primary analysis because low accuracy suggested that they may not have understood or followed the task instructions. This removed 3 participants and left 11 participants in the primary analysis: 8 non ADHD controls and 3 participants with self reported ADHD.

Task design

The experiment used a computerized Picture Word Stroop task adapted from prior modified Stroop paradigms used to study selective attention and interference control (Gernsbacher & Faust, 1991; Hiatt et al., 2004; Newman et al., 1997; Vitale et al., 2007). Each trial required participants to attend selectively to one dimension of a context display while ignoring another. The task included two attention modes. In picture attend trials, participants were instructed to focus on the picture and ignore the word. In word attend trials, participants were instructed to focus on the word and ignore the picture.

Each context display contained a picture and a superimposed word. The picture and word were conceptually unrelated to one another. After the context display, participants saw a single test stimulus, which was either a picture or a word. Participants had to decide whether the test stimulus was conceptually related to the attended part of the earlier context display. They pressed F for “related” and J for “unrelated.” Reaction time and accuracy were recorded for each trial.

The task included three trial types: filler trials, comparison trials, and experimental interference trials. Filler trials required a “related” response because the attended component of the context display was related to the test stimulus. These trials

were included so that participants could not simply respond “unrelated” on every trial. Comparison trials required an “unrelated” response, and neither the attended nor ignored component was related to the test stimulus. These trials served as the baseline for response speed and accuracy when there was no meaningful distractor conflict. Experimental interference trials also required an “unrelated” response, but the ignored component was conceptually related to the test stimulus. These trials measured interference because the correct answer was based on the attended component, while the ignored component supported the opposite response.

The main dependent variables were reaction time and accuracy. The primary reaction time interference score was calculated as mean reaction time on experimental trials minus mean reaction time on comparison trials. A positive interference score indicated slower responses on experimental trials than comparison trials. The primary accuracy interference score was calculated as comparison accuracy minus experimental accuracy. A positive accuracy loss score indicated lower accuracy on experimental trials than comparison trials.

Procedure

Participants completed the task in a standard web browser. At the beginning of the experiment, they entered or received a participant identifier and reported ADHD status. They then read on screen instructions explaining the difference between attending to pictures and attending to words. Participants were told to respond as quickly and accurately as possible.

Each trial began with an attention cue. A “P” cue indicated that participants should attend to the picture in the upcoming context display, while a “W” cue indicated that participants should attend to the word. After the cue, participants saw the context display containing both a picture and a superimposed word. The task then presented a short delay before the test stimulus. The experiment included two interstimulus interval conditions, 300 ms and 750 ms, to vary the amount of time between the context display and the test stimulus. The test stimulus remained on screen until the participant

responded or until the response window ended. The task recorded the participant's response, whether the response was correct, and the reaction time in milliseconds.

Participants first completed 6 practice trials to become familiar with the response keys and task structure. They then completed 15 main trials. The main trials were randomized and included equal numbers of filler, comparison, and experimental trials. Practice trials were not included in the main statistical analyses.

Figure 1. Trial structure of the Picture Word Stroop task

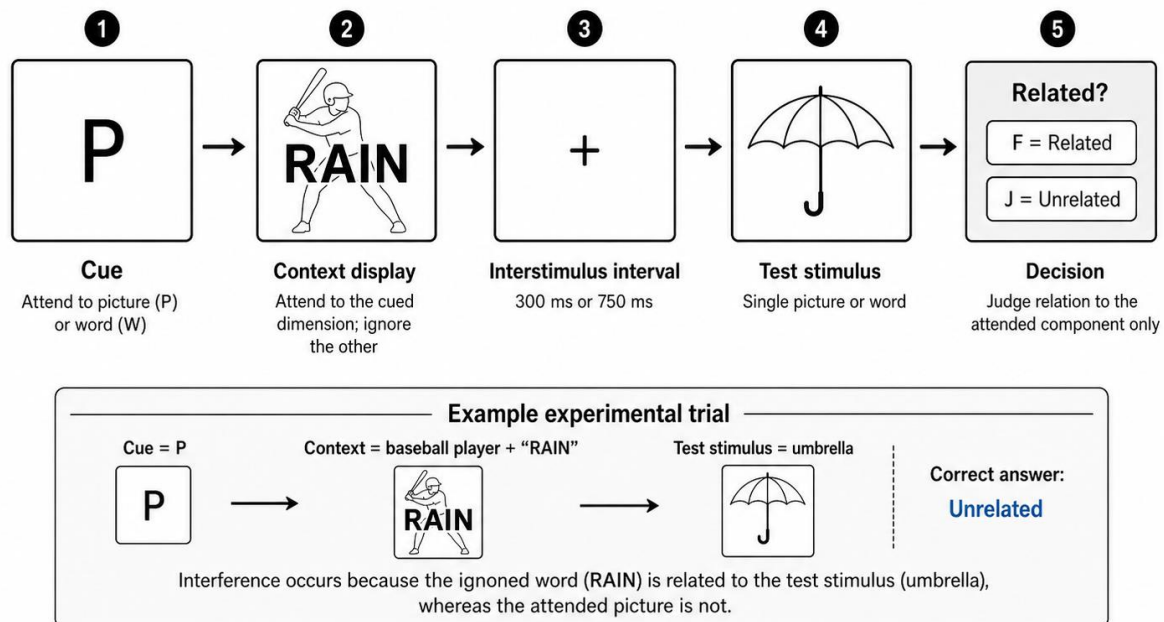


Figure 1. Trial structure of the Picture Word Stroop task. Each trial began with a cue instructing participants to attend either to the picture or to the word. Participants then viewed a context display containing both a picture and a superimposed word. After a short delay, a test stimulus appeared, and participants judged whether it was related to the attended component of the context display. Experimental trials created interference because the ignored component was related to the test stimulus while the attended component was not.

Data processing

Data cleaning was completed before statistical analysis. First, exact duplicate rows were removed from the raw spreadsheet. Second, practice trials were excluded so that only main task performance was analyzed. Third, the participant who selected “prefer not to say” for ADHD status was excluded from group comparisons. Fourth, participants whose main task accuracy was below 75 percent were excluded from the

primary analysis. This was done because very low accuracy suggested that the participant may not have understood the task or may not have been responding carefully.

Accuracy was recalculated directly from the response columns. A trial was coded as correct if the participant response matched the correct response and incorrect if it did not match, including no response trials. For reaction time analyses, only correct trials with an actual response were included. Trials with “no response” were excluded from reaction time analyses because the recorded value represented a timeout rather than a true decision time. Reaction times below 200 ms were also excluded because they were too fast to reflect meaningful stimulus processing. This removed one implausibly fast response from the group analysis.

Statistical analysis

Because the final sample was small and group sizes were unequal, the primary analyses used participant level means rather than trial level means. For each participant, mean reaction time and accuracy were calculated separately for experimental and comparison trials. Reaction time interference was calculated as experimental mean reaction time minus comparison mean reaction time. Accuracy loss was calculated as comparison accuracy minus experimental accuracy.

To test whether the task produced an overall interference effect, one sample t tests compared participant level interference scores against zero. To test the main hypothesis that ADHD participants would show larger interference than controls, Welch independent samples t tests compared the ADHD and non ADHD groups on reaction time interference scores and accuracy loss scores. Welch tests were used because the group sizes were unequal. Effect sizes were reported as Cohen’s d_z for one sample tests and Hedges’ g for between group comparisons. Ninety five percent confidence intervals were included for the main effects. Because of the small sample size, all statistical results were interpreted cautiously.

Results and Discussion (Written by: Soumil Kalra; edited by: Grace Borgerding and Yutong Han)

Data cleaning and participant inclusion

The raw dataset contained 399 rows. After removing exact duplicate rows, the dataset contained 315 unique rows from 15 participants. Each participant completed 6 practice trials and 15 main trials. Practice trials were excluded from the final analysis. The participant who selected “prefer not to say” for ADHD status was excluded from group comparisons because the hypothesis specifically compared ADHD and non ADHD participants. Three additional participants were excluded from the primary analysis because their main task accuracy fell below 75 percent. The final primary analysis included 11 participants: 8 non ADHD controls and 3 participants with self reported ADHD.

This cleaning step was important because the raw dataset contained several signs of noise, including duplicated participant blocks, no response trials, and a small number of unrealistically fast responses. These issues are common risks in short online behavioral tasks, especially when participants complete the experiment without in person supervision. Removing duplicate rows and excluding low accuracy participants made the final analysis more conservative.

Reaction time results

The primary reaction time analysis tested whether participants were slower on experimental interference trials than on comparison trials. Across all valid participants, the mean reaction time interference score was 5.21 ms, meaning that experimental trials were only slightly slower than comparison trials. This overall interference effect was not statistically significant, $t(10) = 0.05$, $p = .959$, Cohen’s $d_z = 0.02$, 95 percent CI [negative 214.31, 224.73]. Therefore, the task did not produce a reliable overall reaction time interference effect in the cleaned sample.

Table 1
Mean reaction time and accuracy by ADHD group in the primary cleaned sample

Group	<i>n</i>	Comparison RT <i>M</i> (SD)	Experimental RT <i>M</i> (SD)	RT interference <i>M</i> (SD)	Comparison accuracy <i>M</i> (SD)	Experimental accuracy <i>M</i> (SD)	Accuracy loss <i>M</i> (SD)
Non ADHD	8	1369.04 (416.95)	1318.79 (386.73)	-50.24 (218.25)	0.98 (0.07)	0.95 (0.09)	0.03 (0.13)
ADHD	3	1636.47 (187.17)	1789.54 (422.20)	153.08 (567.49)	0.87 (0.12)	0.80 (0.20)	0.07 (0.23)

Note. RT = reaction time in milliseconds. Reaction time interference was calculated as experimental RT minus comparison RT. Accuracy loss was calculated as comparison accuracy minus experimental accuracy. Positive values indicate worse performance on experimental trials.

The ADHD group showed a descriptively larger reaction time interference score than the non-ADHD group. Participants with self-reported ADHD had a mean reaction time interference score of 153.08 ms, while non-ADHD controls had a mean score of negative 50.24 ms. This means that, descriptively, ADHD participants were slower on experimental trials than comparison trials, while controls were slightly faster on experimental trials than comparison trials. However, this group difference was not statistically significant, Welch's $t(2.23) = 0.60$, $p = .602$, Hedges' $g = 0.56$, 95 percent CI [negative 1112.77, 1519.41]. Because the confidence interval was extremely wide and included zero, the reaction time data do not provide strong evidence that ADHD participants showed greater interference than controls.

Figure 2. Reaction time interference by ADHD group

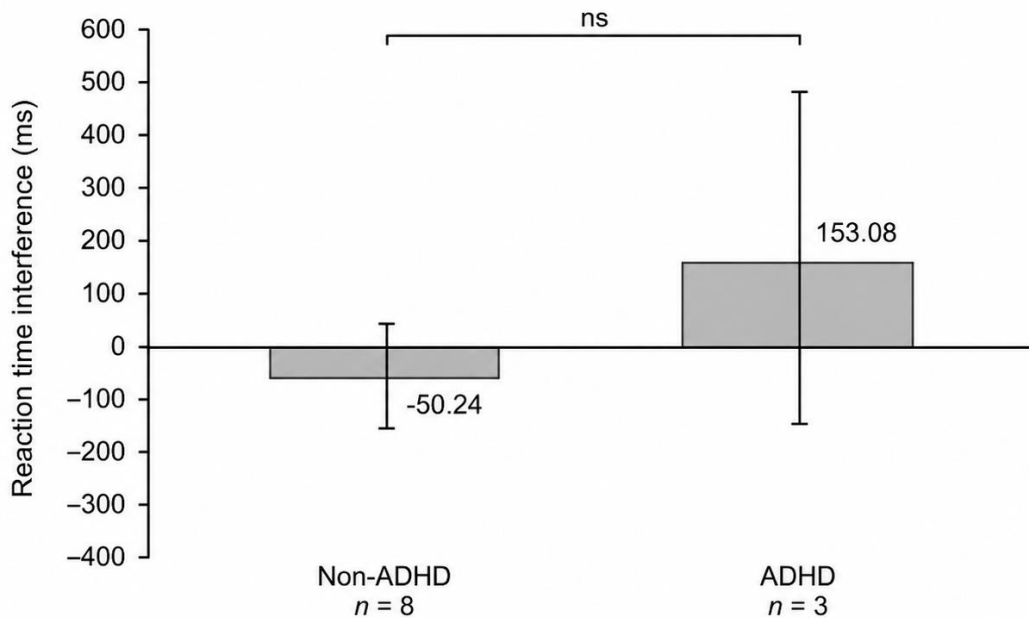


Figure 2. Reaction time interference by ADHD group. Mean reaction time interference scores are shown for the non-ADHD and ADHD groups. Interference was calculated as mean reaction time on experimental trials minus mean reaction time on comparison trials. Positive values indicate slower responses on experimental trials. Error bars represent standard error of the mean. The ADHD group showed a descriptively larger interference score, but the group difference was not statistically significant.

Accuracy results

Accuracy showed a small descriptive pattern in the predicted direction, but the effect was not statistically significant. Across all valid participants, mean accuracy loss was 0.04, which basically means that participants were about 3.64 percentage points less accurate on experimental trials than comparison trials. This overall accuracy interference effect was not statistically significant, $t(10) = 0.80$, $p = .441$, Cohen's $d_z = 0.24$, 95 percent CI [negative 0.06, 0.14].

The ADHD group showed a slightly larger descriptive accuracy loss than the non ADHD group. ADHD participants had an average accuracy loss of 0.07, while non ADHD controls had an average accuracy loss of 0.03. However, this group difference was also not statistically significant, Welch's $t(2.48) = 0.30$, $p = .790$, Hedges' $g = 0.24$, 95 percent CI [negative 0.46, 0.55]. Therefore, the accuracy data also did not provide

statistically reliable evidence that ADHD participants experienced greater interference than controls.

Figure 3. Accuracy loss by ADHD group

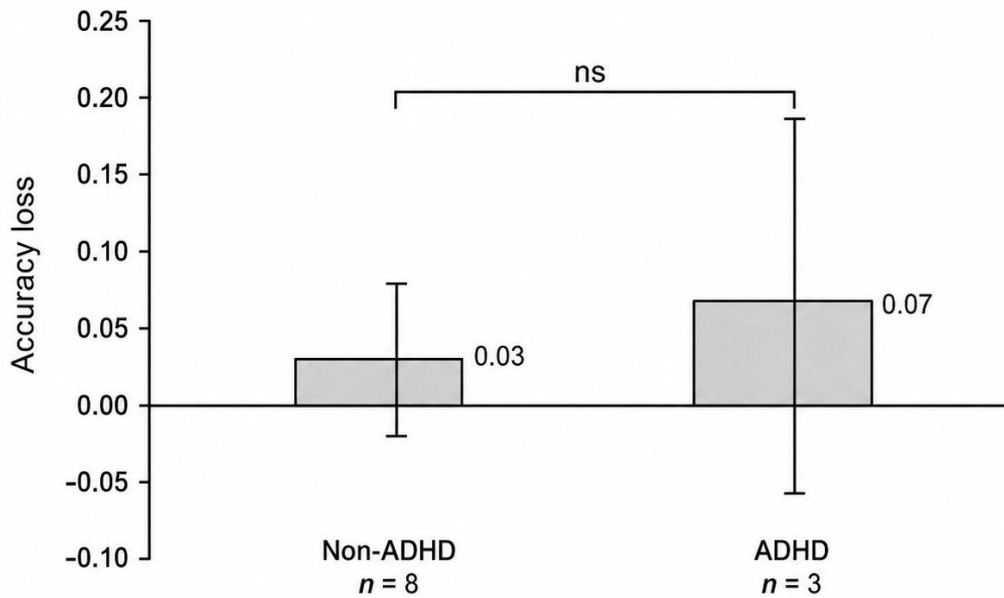


Figure 3. Accuracy loss by ADHD group. Mean accuracy loss scores are shown for the non-ADHD and ADHD groups. Accuracy loss was calculated as comparison accuracy minus experimental accuracy. Positive values indicate lower accuracy on experimental trials. Error bars represent standard error of the mean. The ADHD group showed a slightly larger descriptive accuracy loss, but the group difference was not statistically significant.

Interpretation of the hypothesis

The prediction only had moderate support on the descriptive level, and no statistical support. It was predicted that self-reported ADHD participants would have a larger interference effect than non ADHD controls, and that they would respond more slowly and accurately on experimental trials than on comparison trials. When analyzing the clean reaction times we found a small difference between the groups in the expected direction descriptively (ADHD participants responded more slowly to experimental than to comparison trials). This difference was not statistically significant.

We also found a small difference between the groups in the expected direction for accuracy, but this difference was also not statistically significant.

The most suitable conclusion, then, is that this study offered no conclusive evidence for increased interference control interference with pictures and words in ADHD. However, this does not necessarily mean that ADHD has no involvement in interference control: past studies have shown that people with ADHD have impaired sustained attention, impulse control, and inhibition of distractors (Barkley, 1997; Klein et al., 2025; Rubia, 2018). It is likely that null findings were observed in this study because of some of its design weaknesses, including: a low participant sample size, unequal sample sizes in the groups, a short task length, and messy online data. With only 3 participants in the cleaned, primary analysis that had ADHD, there was incredibly low statistical power in the study to detect a difference in groups. Wide confidence intervals for the group comparison demonstrate that the measure of the difference between groups was too variable to make a definite claim.

Another important issue is that the task may not have included enough trials to produce a stable interference score. Classic Stroop and modified Stroop studies often use many more trials than the 15 main trials used here (Hiatt et al., 2004; MacLeod, 1991; Newman et al., 1997; Vitale et al., 2007). In the present task, each participant completed only 5 experimental trials and 5 comparison trials in the main phase. Because the interference score depended on a small number of trials, a single slow response, mistake, or missed response could strongly influence the participant's average. This is especially important in a task where reaction time is the main outcome.

Comparison with psychopathy literature

Although direct data was not obtained from psychopaths in this study, it is relevant to the existing psychopathy literature. On modified Stroop tasks, it has been shown that psychopaths have an impaired distractor effect particularly on peripheral items, those not relevant to the currently attended information (Hiatt et al. 2004; Newman et al. 1997; Vitale et al. 2007). It has been argued that this suggests an

abnormality in response modulation whereby attentional processes are narrowed to task relevant information and fail to include secondary context.

Our original theoretical framing contrasted this psychopathy pattern with ADHD. If psychopathy reflects reduced processing of peripheral distractors, ADHD might be expected to show the opposite pattern: stronger distractor processing and greater interference. The present data did not provide statistically reliable support for that contrast. However, the framework remains theoretically useful. ADHD and psychopathy may both involve atypical attention, but in different directions. ADHD may involve unstable goal maintenance and distractibility, while psychopathy may involve overfocused attention that fails to incorporate peripheral information. A stronger future study could directly compare ADHD, psychopathy trait measures, and control participants on the same Picture Word Stroop task.

Limitations

The main limitation of this study was sample size. The final primary analysis included only 11 participants, with only 3 in the ADHD group. This made it difficult to detect meaningful group differences and produced wide confidence intervals. A second limitation was that ADHD status was self reported. Participants were not clinically assessed, and no standardized ADHD symptom scale was used. This means that the ADHD group may have varied in symptom severity, medication status, and diagnostic history.

A third factor that limited the study was the small number of trials conducted. Each participant only completed 15 main trials that included 5 experimental trials, 5 comparison trials and 5 filler trials. Although this helped to keep the task brief and therefore less likely to cause fatigue, it compromised the reliability of the interference score. In future adaptations of the task, the number of trials within each condition could be increased but the task kept relatively brief to ensure fatigue was minimized. The final limitation was that the task was run online which resulted in no control over any of the distracting influences on the participant as well as differences in the device used, screen size and participant effort. The last limitation of the task was that there was no

direct psychopath group and thus all references to research in psychopathy must be considered theoretical comparisons rather than empirical ones.

Future directions

Future studies should recruit a larger and more balanced sample, ideally with equal numbers of ADHD and non ADHD participants. The task should include more trials per condition so that reaction time and accuracy estimates are more stable. ADHD status should be measured using a standardized symptom scale, and future research could also collect information about medication use, sleep, and current attentional state. A future version could also include a psychopathy trait questionnaire to test whether ADHD symptoms and psychopathy traits predict opposite patterns of interference within the same task.

A stronger design could also examine the effect of interstimulus interval more directly. The current task included 300 ms and 750 ms ISI conditions, but the sample size was too small to analyze this factor meaningfully. In theory, shorter intervals may capture more automatic distractor processing, while longer intervals may allow more time for suppression. Future studies with more trials could test whether ADHD related interference is stronger at shorter ISIs, which would support the idea that ADHD involves difficulty suppressing distractors during early stages of processing.

Neuroscience Integration (Written by: Grace Borgerding and Soumil Kalra; edited Yutong Han)

The behavioral logic of this study is closely connected to neural models of cognitive control. In the Picture Word Stroop task, participants must maintain a task goal, either attend to the picture or attend to the word, while ignoring a competing stimulus. This requires top down control, which is commonly associated with prefrontal systems that maintain goal relevant information and bias attention toward the correct response pathway (Miller & Cohen, 2001). When the ignored stimulus is conceptually related to the test item, conflict occurs because the distractor activates a competing

response. Conflict monitoring theories suggest that the anterior cingulate cortex helps detect this type of conflict and signals the need for increased control (Botvinick et al., 2001). Therefore, even though our study was behavioral, the task was designed around a neural problem: how the brain maintains a goal while suppressing distracting information.

Our hypothesis is particularly relevant to the framework in the sense that the diagnosis has been linked to abnormalities in activation in networks related to inhibition, conflict monitoring and attention control. Thus studies with fMRI on Stroop-like interference tasks found evidence for anterior cingulate abnormalities in the task with people with ADHD during conflict processing (Bush et al., 1999). Similarly others studies showed reduced activation in the frontal cortex with people with ADHD on executive tasks, especially ones involving higher order control and inhibition (Rubia et al., 1999). Overall, reviews in the field of the neuroscience of ADHD suggest that ADHD consists of problems in the frontostriatal, frontoparietal and cerebellar networks that mediate control of attention and executive function (Rubia, 2018). These findings align with the logic of our hypothesis: a deficient or unstable top down control should render individuals with ADHD more susceptible to non-task-relevant distraction.

Another important neural system is the default mode network. The default mode network includes regions such as the posterior cingulate cortex and precuneus and is usually more active during internally focused thought than externally focused tasks. During demanding attention tasks, healthy participants typically suppress default mode network activity. ADHD has been linked to weaker suppression of this network during task performance, which may contribute to mind wandering and distractibility (Castellanos et al., 2008; Fassbender et al., 2009). In the context of our task, failure to suppress the default mode network could make it harder for participants with ADHD to maintain the current task rule, especially when the ignored word or picture creates a tempting competing response.

The right inferior frontal cortex is also relevant because it is often associated with inhibition and stopping inappropriate responses (Aron et al., 2014). In the Picture Word

Stroop task, participants may need to inhibit the response suggested by the ignored stimulus. For example, if the ignored word is related to the test stimulus, the participant may feel pulled toward a “related” response even though the correct response is “unrelated.” A participant with weaker inhibitory control may therefore show slower responses or lower accuracy on experimental trials. Although our behavioral results did not show a significant ADHD related interference effect, the task still maps clearly onto neural systems involved in inhibition, conflict monitoring, and goal maintenance.

The present results also show why neuroscience integration matters for interpreting null behavioral findings. The lack of significant reaction time or accuracy differences does not necessarily mean that ADHD participants were using the same neural strategies as controls. It is possible that groups could produce similar behavioral performance through different neural mechanisms, especially in a short and relatively easy version of the task. For example, ADHD participants might recruit additional control resources to maintain performance, or the task may not have been difficult enough to reveal behavioral differences. A future fMRI version could test whether ADHD participants show reduced activation in anterior cingulate or inferior frontal regions during experimental trials. An EEG version could test whether ADHD participants show altered timing of distractor suppression, including reduced distractor positivity, which has been discussed as a marker of active distractor filtering (Klein et al., 2025).

In future work, combining this task with neuroimaging would allow a stronger test of the attentional control hypothesis. If ADHD participants show larger behavioral interference, this could be linked to reduced recruitment of prefrontal and anterior cingulate control regions. If ADHD participants do not show larger behavioral interference but show different brain activation patterns, this would suggest compensatory neural processing. Either outcome would be informative. The current study should therefore be seen as a behavioral pilot that establishes the task structure and reveals the practical challenges that would need to be solved before using the task in a larger neuroscience study.

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Contributor Block

Programming: Soumil Kalra was primarily responsible for coding and hosting the web based Picture Word Stroop task. Grace Borgerding and Yutong Han contributed to task planning, debugging feedback, and stimulus/task design decisions.

Presentation: Mainly Grace Borgerding and Yutong Han contributed to the presentation. Soumil Kalra contributed by making the slides for the task demo. Grace Borgerding contributed to the background and neuroscience framing. Yutong Han contributed to the methods and expected results framing.

Abstract: Grace Borgerding and Soumil Kalra wrote the abstract. Yutong Han edited for clarity.

Introduction: Grace Borgerding wrote the original introduction. Soumil Kalra edited and expanded the hypothesis, task framing, and ADHD comparison. Yutong Han edited for flow.

Methods: Soumil Kalra wrote the methods section. Grace Borgerding and Yutong Han edited for clarity and consistency.

Results & Discussion: Soumil Kalra analyzed and wrote the results and discussion. Grace Borgerding and Yutong Han edited the interpretation and connection to the broader literature.

Neuroscience Integration: Grace Borgerding wrote the original neuroscience extension. Soumil Kalra and Yutong Han edited the section to connect it more directly to the behavioral findings and course concepts.

References: Grace Borgerding, Soumil Kalra and Yutong Han compiled the references.